

Jacobsthal Slide and Shiftless Orbit in Collatz Trajectory

Hiroshi Harada — August 1, 2026

Table 1. Shiftless Collatz Orbit of Initial Value 7

Level	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3
J2(0,1)	-5461/16384	2731/8192	-1365/4096	683/2048	-341/1024	171/512	-85/256	43/128	-21/64	11/32	-5/16	3/8	-1/4	1/2	0	1	1	3

Level	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3
J2 Expansion	-2727/8192	1369/4096	-679/2048	345/1024	-167/512	89/256	-39/128	25/64	-7/32	9/16	1/8	5/4	3/2	4	7	15	29	59
Marge	11/8192	11/4096	11/2048	11/1024	11/512	11/256	11/128	11/64	11/32	11/16	11/8	11/4	11/2	11	22	44	88	176
J2 Slide	1371/4096	-677/2048	347/1024	-165/512	91/256	-37/128	27/64	-5/32	11/16	3/8	7/4	5/2	6	11	23	45	91	181
Marge	17/4096	17/2048	17/1024	17/512	17/256	17/128	17/64	17/32	17/16	17/8	17/4	17/2	17	34	68	136	272	544
J2 Slide	-337/1024	175/512	-81/256	47/128	-17/64	15/32	-1/16	7/8	3/4	5/2	4	9	17	35	69	139	277	555
Marge	13/1024	13/512	13/256	13/128	13/64	13/32	13/16	13/8	13/4	13/2	13	26	52	104	208	416	832	1664
J2 Slide	-41/128	23/64	-9/32	7/16	-1/8	3/4	1/2	2	3	7	13	27	53	107	213	427	853	1707
Marge	5/128	5/64	5/32	5/16	5/8	5/4	5/2	5	10	20	40	80	160	320	640	1280	2560	5120
J2 Slide	3/8	-1/4	1/2	0	1	1	3	5	11	21	43	85	171	341	683	1365	2731	5461
Marge	1/8	1/4	1/2	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384
J2 Slide	1/2	0	1	1	3	5	11	21	43	85	171	341	683	1365	2731	5461	10923	21845

Level: The hierarchical shift index. Level 0 represents the fundamental hierarchy of the initial value.

J2(0,1): The basic 2-adic Jacobsthal (J) sequence at each Level, serving as the constant superposition base.

J2 Expansion (Wave): The J-spinor wave to which the current odd kernel belongs.

Marge (Adjacent Sum): The wave interference step. It extracts the pure shift stream by summing adjacent components of the current wave.

J2 Slide: The wave switching step. It generates the exact wave for the next odd Collatz kernel by adding the J2(0,1) base to the Marge.

This matrix visualizes the computational output of the Shiftless Model for the initial value 7.

Reading horizontally reveals the mechanism of wave interference and sliding (J Slide) that governs the Collatz mapping.

Reading vertically down the Level 0 column reveals the pure Shiftless Orbit: $7 \rightarrow 23 \rightarrow 69 \rightarrow 213 \rightarrow 683 \rightarrow 2731$, where the trajectory advances while maintaining its hierarchy, completely bypassing the unpredictable division-by-2 operations.

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